

★ WEEK 6 · PART 3 · PRACTICE

IRF and FEVD, Worked by Hand

Trace a Shock Through the System, Then Budget the Variance

Cholesky, the impulse response, and the variance decomposition

Part 3's tools, with the arithmetic shown. We orthogonalize correlated shocks with a Cholesky factor and trace an impulse response through the VAR, then decompose a variable's forecast-error variance into shares - all while watching how the variable ordering changes the answer.

3P

PART 3 · APPLICATIONS

Shocks, Responses, and Variance

Case A: orthogonalize the correlated shocks with a Cholesky factor, then trace the impulse response across horizons. Case B: decompose the forecast-error variance into shares, and see how the variable ordering changes both. Then a Claude Code lab.

What Part 3 Told Us

Now we trace it - with numbers

1

Shocks are correlated, so orthogonalize

You can't shock one variable alone when shocks move together. A Cholesky factor turns them into independent unit shocks.

2

The IRF traces a shock through time

It's the response of each variable to an orthogonalized shock at every horizon - the system's dynamic story, decaying if stable.

3

The FEVD budgets the uncertainty

It splits each variable's forecast-error variance into shares from each shock - and both it and the IRF depend on the ordering.

Everything here is a calculator exercise: one Cholesky factor, one impulse response, one variance budget.

How These Cases Work

Data, then arithmetic, then meaning

Each case has three steps

- Start from the shock covariance and matrix.
- Trace the response, or split the variance.
- Then read the dynamics.

What you should watch

- How the Cholesky untangles the shocks.
- How the response decays over horizons.
- How ordering changes both answers.

Orthogonalize, Then Trace

A shock's path through the system

A stock's and its sector's shocks move together, so a shock to one already implies a move in the other. To interpret a clean shock, we orthogonalize with a Cholesky factor, then push a unit shock through the VAR to trace the impulse response over months. We compute the factor and the first few horizons by hand

The setup

- Shock covariance with correlation 0.4.
- The VAR matrix from Part 1.
- A unit shock to variable 1.

What we'll do by hand

- Compute the Cholesky factor.
- Read the impact response.
- Iterate the matrix for later horizons.

Orthogonalization

Untangle the correlated shocks

$$\Sigma = P P^{\top} \text{ (lower-triangular } P) : \text{ correlated shocks } \longrightarrow \text{ independent unit shocks}$$

The problem

- Shocks are correlated, so they co-move.
- Shocking one alone isn't well-defined.
- Interpretation needs independent shocks.

The fix

- Factor the covariance as P times P -transpose.
- P is lower-triangular - the Cholesky.
- Its columns are independent unit shocks.

The Cholesky Factor

A triangular square root

$$\Sigma = \begin{bmatrix} 1.0 & 0.4 \\ 0.4 & 1.0 \end{bmatrix} \Rightarrow P = \begin{bmatrix} 1.0 & 0 \\ 0.4 & 0.917 \end{bmatrix} \quad (P_{22} = \sqrt{1 - 0.4^2})$$

The entries

- Top-left is the square root of 1.0.
- Bottom-left is 0.4 - the correlation.
- Bottom-right is root of one minus 0.16.

What it means

- A shock to var 1 hits var 1 by 1.0.
- And var 2 by 0.4 - the shared part.
- That's the orthogonalized impact.

The Impulse Response

Push the shock through the matrix

$$\text{IRF}(h) = A^h P e_1 : \begin{bmatrix} 1.00 \\ 0.40 \end{bmatrix} \rightarrow \begin{bmatrix} 0.62 \\ 0.26 \end{bmatrix} \rightarrow \begin{bmatrix} 0.39 \\ 0.17 \end{bmatrix} \text{ (decays)}$$

The impact response is $[1.00, 0.40]$. Multiplying by the VAR matrix gives the next horizon $[0.62, 0.26]$, then $[0.39, 0.17]$ - the shock propagates and decays, because the VAR is stable.

The Shock Ripples, Then Fades

A one-unit orthogonalized shock to variable 1 hits it at 1.0 and the sector at 0.4, then both responses decay smoothly toward zero over the following months. The decay is guaranteed by the VAR's stability - shocks die out rather than compounding

Case A: a unit shock to var 1 propagates, then decays to zero (stable VAR)



Why Orthogonalize First

Clean shocks, honest stories

Without orthogonalization

- A shock to one variable is ill-defined.
- You'd double-count the shared movement.
- The response would be uninterpretable.

With the Cholesky

- Each shock is independent and unit-sized.
- The response traces a clean causal-style story.
- Horizons show how long the effect lasts.

Budget the Variance

Where does the uncertainty come from?
Beyond the shape of a response, we want to know how much of a variable's forecast uncertainty each shock explains. The FEVD splits the forecast-error variance into shares that sum to one, horizon by horizon. We compute variable 1's shares - and then see how reordering the variables changes both the Cholesky and the

The question

- How much of var 1's forecast error is its own?
- How much comes from var 2's shock?
- And how does that shift with the horizon?

SW

What we'll do by hand

- Sum the squared responses per shock.
- Divide by the total for the shares.
- Then swap the ordering.

The FEVD

A share of forecast-error variance

$$\text{FEVD}_{ij}(h) = \frac{\text{variance of } i \text{ from shock } j \text{ up to } h}{\text{total forecast-error variance of } i} \quad (\text{shares sum to } 1)$$

How it's built

- Square the responses to each shock.
- Sum them up to the horizon.
- Divide by the grand total.

How to read it

- Each share is a slice of the variance.
- They add to one hundred percent.
- A variance budget, horizon by horizon.

The Variance Shares

Var 1's budget, horizon by horizon

horizon	own shock (var 1)	other shock (var 2)
$h = 1$	100%	0%
$h = 3$	92%	8%
$h = 6$	90%	10%

At one step, all of variable 1's forecast error is its own shock. As the horizon grows, the sector's shock seeps in - reaching about 10% by six months - because the VAR lets the two series interact over time.

The Ordering Problem

A choice that changes the answer

order [1, 2] : var 1 absorbs the shared shock $\neq$ order [2, 1] : var 2 does (answer changes)

Why it happens

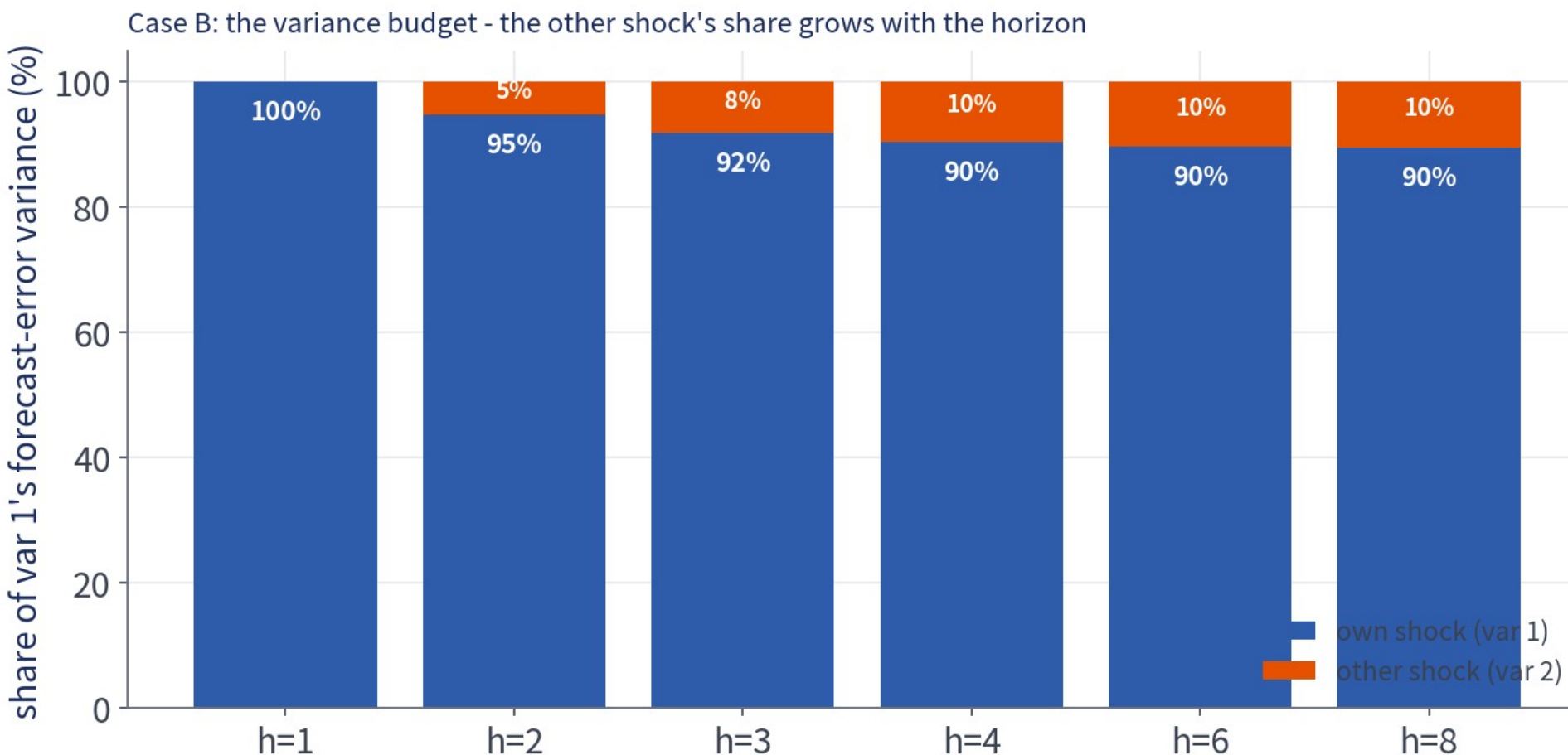
- The Cholesky is triangular.
- The first-ordered variable takes the shared shock.
- So its impact is always the full unit.

What to do

- Order by economic priority - most exogenous first.
- Report how sensitive results are to the order.
- Or use ordering-free alternatives.

The Variance Budget Over Time

At the one-step horizon, variable 1's forecast error is entirely its own shock. As the horizon lengthens, the orange slice - the sector's shock - grows to about a tenth. The stacked bars always sum to 100%: every unit of uncertainty is attributed to some shock



A Budget, With a Warning

Useful, but order-dependent

What the FEVD gives you

- Which shocks drive a variable's uncertainty.
- How that mix shifts across horizons.
- A clear attribution of forecast risk.

The warning

- Both IRF and FEVD depend on the ordering.
- Change the order, change the numbers.
- State your ordering and defend it.

Three Lenses on a VAR

Granger Causality and IR/ID together

The week's three tools answer three questions about the same system. Granger causality asks whether one series helps predict another. The impulse response asks how a shock propagates and how long it lasts. The variance decomposition asks how much of each variable's uncertainty each shock explains. Orthogonalization

The response (Case A)

- Cholesky factor $[[1,0],[0.4,0.917]]$.
- Impact $[1.0, 0.4]$, decaying after.
- A clean, orthogonalized shock.

The budget (Case B)

- Own shock 100% at $h=1$, 90% at $h=6$.
- Sector's share grows with the horizon.
- But the ordering sets it all.

Reproduce the Response and the Budget

Direct Claude Code to check your arithmetic

GOAL Orthogonalize, trace the IRF, and compute the FEVD - then swap the ordering.

[INPUT] Prompt · Claude Code

```
Write and run part3_practice.py that, with
A=[[0.5,0.3],[0.1,0.4]] and Sigma=[[1,0.4],[0.4,1]]:
CASE A - IRF:
    * print the Cholesky factor P;
    * trace the IRF to a var-1 shock for h=0..3.
CASE B - FEVD and ordering:
    * print var 1's FEVD shares at h=1,3,6;
    * recompute P for the reversed ordering and show
      the impact response changes.
Then plot the IRF and the stacked FEVD.
```

Check against the slides

- $P = \begin{bmatrix} 1 & 0 \\ 0.4 & 0.917 \end{bmatrix}$.
- IRF $[1.0, 0.4] \rightarrow [0.62, 0.26]$.
- FEVD 100% $\rightarrow$ 90% own.
- reorder flips who absorbs.

Deliverable: part3_practice.py whose printed numbers match the IRF, the FEVD, and the ordering effect.

Three Questions on the Cases

If you can answer these, the practice has landed

Q1 Why must you orthogonalize before an IRF?

**Correlated shocks
can't be moved one
at a time**

Q2 What does the FEVD tell you?

**Each shock's share of
a variable's forecast
variance**

Q3 Why does the variable ordering matter?

**The Cholesky gives
the shared shock to
the first variable**

All three come from the Cholesky, the impulse response, and the variance decomposition.

Three Pitfalls to Avoid

Where IRF and FEVD trip people

Common mistakes

- "Shock one variable and hold the others fixed."
- "The IRF is the same for any ordering."
- "A FEVD share is a causal contribution."

The reality

- Correlated shocks must be orthogonalized first.
- Ordering changes both the IRF and the FEVD.
- It's a forecast-variance share, not proven cause.

Two Calculations to Remember

Part 3 applications in one screen

1

Orthogonalize, then trace the response

The Cholesky factor of the correlated covariance was $[[1.0, 0], [0.4, 0.917]]$, giving an impact response of $[1.0, 0.4]$. Pushed through the VAR, it decayed to $[0.62, 0.26]$ then $[0.39, 0.17]$ - a clean shock rippling out and fading.

2

Budget the variance, mind the order

Variable 1's forecast-error variance was 100% its own shock at one step, falling to about 90% by six months as the sector's shock seeped in. But reordering the variables changes the Cholesky - and both the IRF and the FEVD with it.

3

Three lenses on one system

Granger causality tests predictability, the IRF traces a shock's path, and the FEVD budgets the uncertainty. Together they turn a fitted VAR into a full picture of how the variables move each other.

Week 6 complete — from the VAR model, through Granger causality, to impulse responses and variance decomposition.